

Exercise 6.4. Let R be a ring. An element $x \in R$ is called *nilpotent* if there exists a positive integer $n \in \mathbb{N}_{>0}$ such that $x^n = 0$. Show that the set of all nilpotent elements in R is an ideal.

You can try to [verify this exercise with Lean](#).

Hints for Lean

```
import Mathlib.RingTheory.Ideal.Basic
import Mathlib.RingTheory.Nilpotent.Basic

variable (R : Type*) [CommRing R]

namespace Ideal

def nilradical : Ideal R :=
{ carrier := {x : R | IsNilpotent x}
  zero_mem' := by
    -- Proof that it contains the zero element.
    sorry
  add_mem' := by
    -- Proof that it is closed under addition.
    sorry
  smul_mem' := by
    -- Proof that it is closed under scalar multiplication.
    sorry
}

end Ideal
```

A [solution](#) is available on the repository

<https://github.com/pedro-nunez/lean-algebra-2021>.

[†]Exercises designed for the lecture *Algebra and Number Theory*, for which I was a teaching assistant. This was an introductory lecture on abstract algebra, polynomial rings and Galois theory, taught by Annette Huber-Klawitter at the University of Freiburg during the Winter Semester 2021/2022.